2015-2 期末考试试卷解答

1-6. CAC BDB　**7.** $3z^2 - y^2 = 16$　**8.** $\{0, x, 0\}$　**9.** $\sqrt[3]{\dfrac{3}{2}}$　**10.** 5

11. 设所求直线的方向矢量为$\mathbf{s}$，与z轴相交于$Q(0,0,c)$，则$\mathbf{s} = \overrightarrow{PQ} = \{-1,-2,c-3\}$，由题设，$\mathbf{s} \perp \{1,1,1\}$，故$c = 6$，所以$\mathbf{s} = \{-1,-2,3\}$，直线方程为$\dfrac{x-1}{1} = \dfrac{y-2}{2} = \dfrac{z=3}{-3}$.

12. $z_x = \cos x f_1 + \dfrac{1}{y} f_2$，　$z_{xy} = -\dfrac{x\cos x}{y^2} f_{12} - \dfrac{1}{y^2} f_2 - \dfrac{x}{y^3} f_{22}$

13. 方程组$\begin{cases} z = xf(x+y) \\ F(x,y,z) = 0 \end{cases}$两边对$x$求导：$\begin{cases} \dfrac{dz}{dx} = f + x(1 + \dfrac{dy}{dx}) f' \\ F_1 + \dfrac{dy}{dx} F_2 + \dfrac{dz}{dx} F_3 = 0 \end{cases}$，所以

$$\frac{dz}{dx} = \frac{\begin{vmatrix} -xf' & f + xf' \\ F_2 & -F_1 \end{vmatrix}}{\begin{vmatrix} -xf' & 1 \\ F_2 & F_3 \end{vmatrix}} = \frac{xf'F_1 - (f + xf')F_2}{F_2 + xf'F_3} \quad （F_2 + xf'F_3 \neq 0）.$$

14. $\iiint\limits_V (x^2 + yz + z^2) dv = \dfrac{2}{3} \iiint\limits_V (x^2 + y^2 + z^2) dv = \dfrac{2}{3} \int_0^{2\pi} d\theta \int_0^{\pi} \sin\varphi d\varphi \int_0^1 \rho^4 d\rho = \dfrac{8\pi}{15}$.

15. 由格林公式 $I = \iint\limits_D (2x + 2y) dxdy \ = 2(\bar{x} + \bar{y})\sigma$（$\sigma$表示区域$D$的面积）$= 3\pi$.

16. （1）令$x - 3 = t$，对于级数$\sum\limits_{n=1}^{\infty} \dfrac{t^n}{n \cdot 3^n}$，$R = \lim\limits_{n\to\infty} \left| \dfrac{a_n}{a_{n+1}} \right| = \lim\limits_{n\to\infty} \dfrac{(n+1)3^{n+1}}{n3^n} = 3$,当$t = 3$时，级数为$\sum\limits_{n=1}^{\infty} \dfrac{1}{n}$发散；当$t = -3$时，级数为$\sum\limits_{n=1}^{\infty} \dfrac{(-1)^n}{n}$收敛，故级数$\sum\limits_{n=1}^{\infty} \dfrac{t^n}{n \cdot 3^n}$的收敛域为$[-3,3)$，原级数的收敛域为$[0,6)$.

$$S(x) = \sum_{n=1}^{\infty} \frac{(x-3)^n}{n \cdot 3^n} = \sum_{n=1}^{\infty} \frac{(\frac{x-3}{3})^n}{n} = \sum_{n=1}^{\infty} \frac{t^n}{n} \quad （令 t = \frac{x-3}{3}）$$

$$= \sum_{n=1}^{\infty} \int_0^t u^{n-1} du = \int_0^t \sum_{n=1}^{\infty} u^{n-1} du = \int_0^t \frac{1}{1-u} du = -\ln(1-t) = -\ln(1 - \frac{x-3}{3}),\quad x \in [0,6) .$$

或者利用已知结果直接写出结论：$S(x) = \sum\limits_{n=1}^{\infty} \dfrac{(x-3)^n}{n \cdot 3^n} = \sum\limits_{n=1}^{\infty} \dfrac{(\frac{x-3}{3})^n}{n} = \ln(1 - \dfrac{x-3}{3})$，$x \in [0,6)$.

17. 设所求点为$M(x,y,z)$，$\nabla f(M)=\{2x,2y,2z\}$， $\mathbf{n}^{\circ}=\frac{1}{\sqrt{2}}\{0,-1,1\}$， $f(x,y,z)$在点$M(x,y,z)$处的方向导数为$\frac{\partial f(M)}{\partial \mathbf{n}}=\sqrt{2}(-y+z)$。构造函数$L(x,y,z,\lambda)=(z-y)+\lambda(x^2+2y^2+2z^2-1)$

令 $\begin{cases} L_x=2\lambda x=0 \\ L_y=-1+4\lambda y=0 \\ L_z=1+4\lambda z=0 \\ L_\lambda=x^2+2y^2+2z^2-1=0 \end{cases}$ $\Rightarrow x=0, y=-z$，代到最后一个方程，得

$y=\pm\frac{1}{2}, z=\mp\frac{1}{2}$，即受检点为$M_1(0,\frac{1}{2},-\frac{1}{2})$， $M_2(0,-\frac{1}{2},\frac{1}{2})$，

因为 $\frac{\partial f(M_1)}{\partial n}=-\sqrt{2}$，$\frac{\partial f(M_2)}{\partial n}=\sqrt{2}$，所以$M_2(0,-\frac{1}{2},\frac{1}{2})$为所求。

18. $\Phi=\iint\limits_S \mathbf{v}\cdot\mathbf{n}dS=\iint\limits_S 2x^3dydz+2y^3dzdx+3(z^2-1)dxdy$. 补$S_1$：$z=0(x^2+y^2\le 1)$取下侧，则$S+S_1$封闭，且指向外侧。所以 $\Phi=\oiint\limits_{S+S_1}-\iint\limits_S=\iiint\limits_V(6x^2+6y^2+6z)dv-\iint\limits_{S_1}3(z^2-1)dxdy$

其中 $\iiint\limits_V(6x^2+6y^2+6z)dv=6\int_0^{2\pi}d\theta\int_0^1 rdr\int_0^{1-r^2}(r^2+z)dz=2\pi$；

$\iint\limits_{S_1}3(z^2-1)dxdy=3\iint\limits_{x^2+y^2\le 1}(-1)(-dxdy)=3\pi$，所以 $\Phi=-\pi$.

19. 参见教材。

20. 因$f(1,y)=0, f(x,1)=0$，所以$f_y(1,y)=0, f_x(x,1)=0$，依据分部积分法，

$$\iint\limits_D xyf_{xy}(x,y)d\sigma=\int_0^1dx\int_0^1 xyf_{xy}(x,y)dy=\int_0^1dx\int_0^1 xyd[f_x(x,y)]$$

$$=\int_0^1[xyf_x(x,y)\Big|_0^1-\int_0^1 xf_x(x,y)dy]dx=-\int_0^1dx\int_0^1 xf_x(x,y)dy$$

$$=-\int_0^1dy\int_0^1 xd[f(x,y)]=-\int_0^1[xf(x,y)\Big|_0^1-\int_0^1 f(x,y)dx]dy=\int_0^1dy\int_0^1 f(x,y)dx=a.$$